

22-Multiple_Linear_Regression

October 22, 2024

1 Multiple Linear Regression

```
[1]: import random
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, root_mean_squared_error, r
↪ r2_score

data = []

for _ in range(500):
    team_name = f"Team {chr(random.randint(65,90))}"
    season = random.randint(2010, 2023)
    wins = random.randint(50, 110)
    losses = 162 - wins
    hits = random.randint(1200,1600)
    doubles = random.randint(200,350)
    triples = random.randint(10,40)
    home_runs = random.randint(100,250)
    strikeouts = random.randint(1000,1500)

    hits_adjusted = hits + (wins-80) * 5
    doubles_adjusted = doubles + (wins-80) * 2
    triples_adjusted = triples + (wins-80) * 3
    home_runs_adjusted = home_runs + (wins-80) * 3
    strikeouts_adjusted = strikeouts - (wins - 80) * 10

    data.append([team_name, season, wins, losses, hits_adjusted, r
↪doubles_adjusted, triples_adjusted, home_runs_adjusted, strikeouts_adjusted])
```

```
[2]: columns = ["Team", 'Season', 'Wins',"Losses", 'Hits', 'Doubles', "Tripples",
↳ 'HomeRuns', ' Strikesouts']
```

```
df = pd.DataFrame(data, columns=columns)
df
```

```
[2]:
```

	Team	Season	Wins	Losses	Hits	Doubles	Tripples	HomeRuns	\
0	Team R	2018	68	94	1516	197	-17	172	
1	Team Y	2023	65	97	1344	275	-6	153	
2	Team U	2017	62	100	1457	197	-26	152	
3	Team U	2011	62	100	1182	233	-28	68	
4	Team N	2016	75	87	1248	233	0	194	
..	...	...	...	...	...	...	...	...	
495	Team D	2010	56	106	1106	152	-40	173	
496	Team O	2022	104	58	1587	310	93	307	
497	Team F	2019	106	56	1612	402	104	285	
498	Team Y	2010	64	98	1320	178	-30	126	
499	Team F	2019	55	107	1090	299	-41	87	

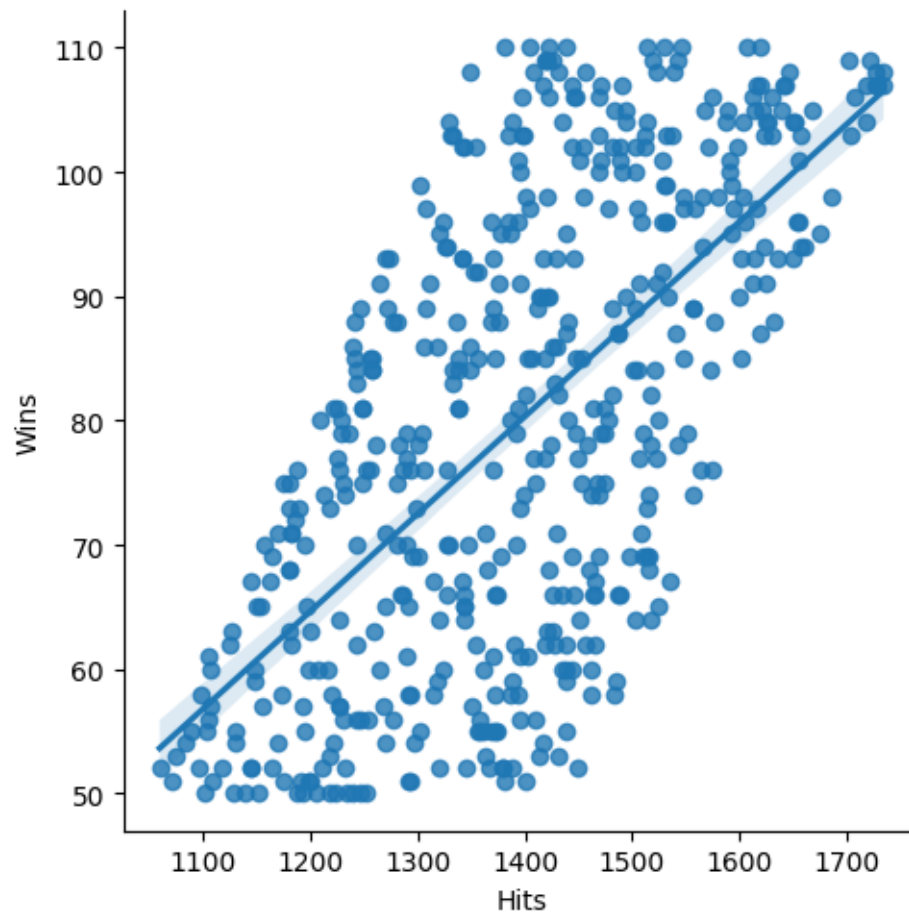
```

Strikesouts
0          1582
1          1495
2          1454
3          1271
4          1058
..          ...
495         1341
496         1130
497         1224
498         1204
499         1573
```

```
[500 rows x 9 columns]
```

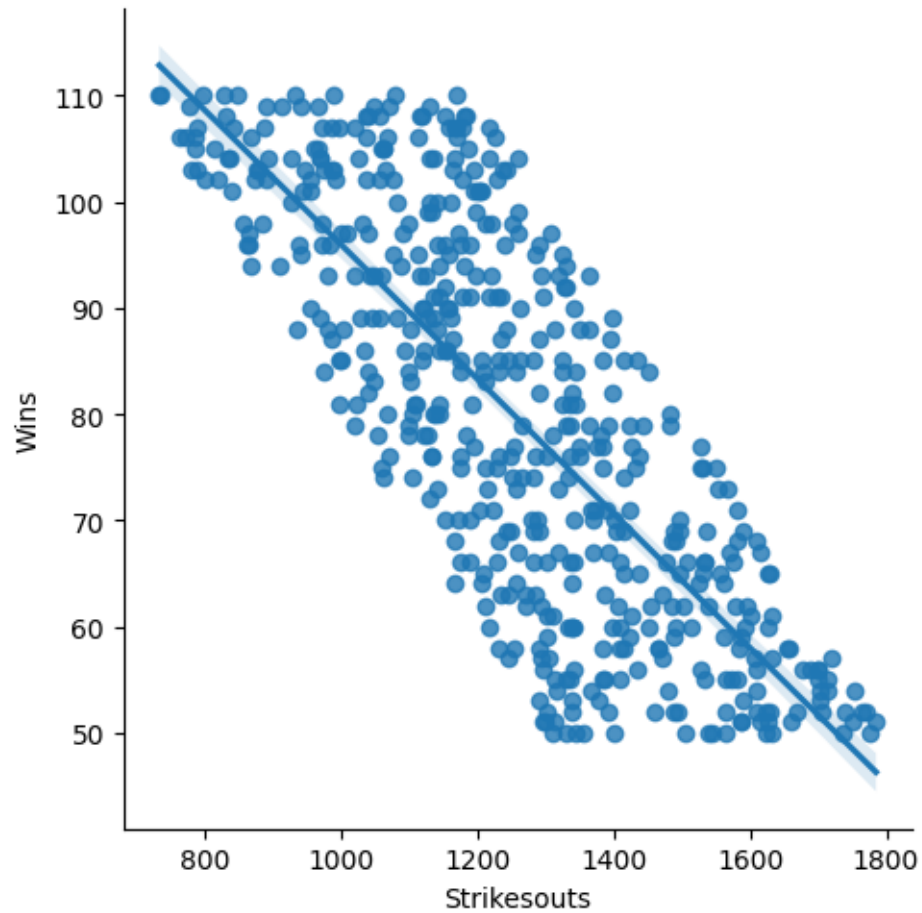
```
[3]: sns.lmplot(x="Hits", y="Wins", data=df)
```

```
[3]: <seaborn.axisgrid.FacetGrid at 0x143f5ca4bd0>
```



```
[4]: sns.lmplot(x="Strikesouts", y="Wins", data=df)
```

```
[4]: <seaborn.axisgrid.FacetGrid at 0x143c0b5b450>
```



```
[5]: df2 = df.drop(columns=['Team', 'Season', 'Losses'], axis=1)
df2
```

```
[5]:
```

	Wins	Hits	Doubles	Tripples	HomeRuns	Strikesouts
0	68	1516	197	-17	172	1582
1	65	1344	275	-6	153	1495
2	62	1457	197	-26	152	1454
3	62	1182	233	-28	68	1271
4	75	1248	233	0	194	1058
..	...	...	...	...	...	...
495	56	1106	152	-40	173	1341
496	104	1587	310	93	307	1130
497	106	1612	402	104	285	1224
498	64	1320	178	-30	126	1204
499	55	1090	299	-41	87	1573

```
[500 rows x 6 columns]
```

```
[6]: df2.columns
```

```
[6]: Index(['Wins', 'Hits', 'Doubles', 'Tripples', 'HomeRuns', 'Strikesouts'],  
      dtype='object')
```

```
[7]: X = df[['Hits', 'Doubles', 'Tripples', 'HomeRuns', 'Strikesouts']]  
     y = df['Wins']  
  
     X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,  
     ↪random_state=24)  
  
     lr = LinearRegression()  
     lr.fit(X_train, y_train)
```

```
[7]: LinearRegression()
```

```
[8]: lr.score(X_test, y_test)
```

```
[8]: 0.9766876951321011
```

```
[9]: lr.score(X_train, y_train)
```

```
[9]: 0.9779524643162276
```

```
[10]: y_pred = lr.predict(X_test)  
      y_pred
```

```
[10]: array([ 79.43521812,  84.98752505,  50.06532199,  86.3822643 ,  
          70.2017775 , 105.95777075, 112.95922339,  63.40753436,  
          69.69225294,  61.45407411, 109.65817756,  91.24687093,  
          53.51602938,  93.85145009,  52.19256777, 103.42618223,  
          93.19739631,  49.75559028,  69.68435674,  66.0810152 ,  
          55.59211481,  55.51175199,  83.75514524,  87.95757516,  
          80.68253472,  89.01173975,  66.41065628,  51.2704802 ,  
         114.21228759, 100.30212302,  88.33733575,  78.18776603,  
          84.57104209,  55.42473415,  68.36036516,  77.72199593,  
          63.81868048,  78.23857521,  95.88171008,  46.68880483,  
         106.20061061,  99.65684269, 102.86836133,  76.59226093,  
          97.43890582,  76.61837152,  67.35962044,  96.54023581,  
          98.99463936,  56.07128436,  73.84535217,  66.34627102,  
         100.5049504 ,  74.42990629,  66.88490465, 103.15534167,  
          97.23129157,  64.7906051 ,  69.02154043,  67.47991565,  
          62.22324859,  57.34789107, 103.81785868,  97.64835468,  
          77.99833509,  98.76947005, 106.69044356,  56.70169242,  
         111.37678841,  69.6311106 ,  99.53710032,  61.06668516,  
         104.96695125,  90.96773068,  96.64482364,  49.51410132,  
          67.4316584 ,  56.55871945, 101.21202627, 103.06747698,  
          88.72028133, 100.99416768,  90.87960032,  95.11780101,
```

```
105.01142639, 77.77341573, 57.13396359, 75.36364628,  
62.00493217, 66.91415242, 72.84128309, 97.89586341,  
109.90864271, 112.59300656, 76.58139883, 89.71476103,  
97.45697597, 80.89370933, 110.05559584, 56.32579865])
```

```
[11]: mean_absolute_error(y_test, y_pred)
```

```
[11]: 2.394583294287179
```

```
[12]: root_mean_squared_error(y_test, y_pred)
```

```
[12]: 2.7715279457881903
```

```
[13]: r2_score(y_test, y_pred)
```

```
[13]: 0.9766876951321011
```

```
[14]: lr.coef_
```

```
[14]: array([ 0.00362517,  0.00364042,  0.29551377,  0.01535465, -0.00312123])
```

```
[15]: lr.intercept_
```

```
[15]: 67.51826690267674
```